Linear Algebra in Post Quantum Cryptography: The Learning with Errors (LWE) Problem

Ethel Adwoa Sakyi
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1 Introduction

The Learning With Errors (LWE) problem is one of the central foundations of modern post-quantum cryptography. Although real LWE systems involve high-dimensional lattices and abstract algebra, the essential behavior can be understood using introductory linear algebra. This paper develops the main ideas through a single consistent example:

$$A = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}, \qquad b = \begin{pmatrix} 5 \\ 10 \end{pmatrix}.$$

The system is first solved over the real numbers, then with modular arithmetic, and then with the noise added. These steps show how the noise changes a simple algebraic problem into a difficult geometric one which is a concept that plays an important role in post-quantum cryptography.

2 Solving the System Over the Real Numbers

Consider the system

$$\begin{cases} 2x_1 + x_2 = 5, \\ 3x_1 + 4x_2 = 10. \end{cases}$$

The augmented matrix is

$$\left[\begin{array}{cc|c} 2 & 1 & 5 \\ 3 & 4 & 10 \end{array}\right]$$

Row reduction proceeds as follows.

Step 1: Normalize the first row

$$R_1 \leftarrow \tfrac{1}{2}R_1 \quad \Rightarrow \quad \left[\begin{array}{cc|c} 1 & \frac{1}{2} & \frac{5}{2} \\ 3 & 4 & 10 \end{array}\right].$$

Step 2: Clear the entry below the pivot

$$R_2 \leftarrow R_2 - 3R_1 \Rightarrow \left[\begin{array}{cc|c} 1 & \frac{1}{2} & \frac{5}{2} \\ 0 & \frac{5}{2} & \frac{5}{2} \end{array}\right].$$

Step 3: Normalize the second pivot

$$R_2 \leftarrow \tfrac{2}{5}R_2 \Rightarrow \left[\begin{array}{cc|c} 1 & \frac{1}{2} & \frac{5}{2} \\ 0 & 1 & 1 \end{array}\right].$$

Step 4: Clear above

$$R_1 \leftarrow R_1 - \tfrac{1}{2}R_2 \Rightarrow \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right].$$

Thus the solution is

$$x = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

3 Choosing a Modulus

To solve a system modulo q, division must be replaced by multiplication by modular inverses. If any pivot in row reduction is non-invertible modulo q, the modulus is unsuitable.
A number a has an inverse modulo 7 if

$$ak \equiv 1 \pmod 7$$

for some $k \in \{1, \ldots, 6\}$. Computation shows that 2, 3, 4, and 6 are all invertible modulo 7, so 7 is a suitable modulus.
In contrast, reducing the system modulo 3 produces the contradictory row

$$[0 \;\; 0 \mid 1],$$

showing that 3 is not a valid modulus for this system.

4 Solving the System Modulo 7

Reducing $10 \equiv 3 \pmod 7$, the augmented matrix becomes $\left[\begin{array}{cc|c} 2 & 1 & 5 \\ 3 & 4 & 3 \end{array}\right]_{\text{mod } 7}$.

Step 1: Normalize the first row

$$R_1 \leftarrow 4R_1 \Rightarrow \left[\begin{array}{cc|c} 1 & 4 & 6 \\ 3 & 4 & 3 \end{array}\right].$$

Step 2: Clear below

$$R_2 \leftarrow R_2 - 3R_1 \Rightarrow \begin{bmatrix} 1 & 4 & 6 \\ 0 & 6 & 6 \end{bmatrix}.$$

Step 3: Normalize the second pivot

$$R_2 \leftarrow 6R_2 \Rightarrow \begin{bmatrix} 1 & 4 & 6 \\ 0 & 1 & 1 \end{bmatrix}.$$

Step 4: Clear above

$$R_1 \leftarrow R_1 - 4R_2 \Rightarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}.$$

Thus modulo 7,

$$x_1 \equiv 2, \qquad x_2 \equiv 1,$$

which matches the real-number solution.

5 Dimension and the Lattice Interpretation

The matrix A is 2×2, so each vector involved has two coordinates. The outputs Ax lie in a two-dimensional space and may be viewed as points on a grid. The set of all such points, as x ranges over integer vectors, forms a lattice.
Without noise, the observed vector lies exactly on a lattice point. With noise, it is displaced slightly off the lattice.

6 Adding Noise

Let

$$e = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \equiv \begin{pmatrix} 1 \\ 6 \end{pmatrix} \pmod 7.$$

Then the noisy vector is

$$b' = \begin{pmatrix} 5 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ 6 \end{pmatrix} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}.$$

Solving $Ax = b' \pmod 7$ produces

$$x_1 \equiv 4, \qquad x_2 \equiv 3,$$

which is not the true solution. Row reduction fails because the vector no longer lies on the lattice.

7 Conclusion

This small example reflects the basic structure of the Learning With Errors (LWE) problem used in post-quantum cryptography. In an actual LWE system, the matrix A is chosen by the legitimate user and made public. The vector x serves as the secret key and is kept private. To protect this secret, a small noise vector is intentionally added to the product Ax. Without noise, an attacker could simply solve the linear system and recover x, but once noise is added, the observed value no longer sits exactly on the lattice generated by A. Standard algebraic techniques then fail to recover the correct solution.

In the two-dimensional example considered here, this behavior is easy to visualize: the noise slightly shifts the point off the lattice, and row reduction no longer returns the original vector. In real LWE-based cryptography, the same idea is expanded to hundreds of dimensions. In such large spaces, determining which lattice point a noisy value came from becomes extremely difficult, even for quantum computers. This difficulty forms the basis of modern post-quantum cryptosystems, which rely on the fact that recovering the secret vector from noisy data is computationally infeasible.

Thus, the elementary computations in this paper illustrate the core idea behind LWE: deliberately adding noise turns an easy algebra problem into a hard geometric one, and scaling this idea to high dimensions produces cryptographic systems designed to remain secure in the quantum era.